Chaowei Fang

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One-Page Summary

Current Position

- Lecturer, Key Laboratory of Intelligent Perception and Image Understanding of Ministry of Education, School of Artificial Intelligence, Xidian University, Xi'an, China

Research Fields and Interests

- Low-level natural/medical image processing; network design; annotation-efficient machine learning

Education

- **The University of Hong Kong** — Hong Kong
  Ph.D. - Computer Science; Advisor: Prof. Yizhou Yu — *09/2015 - 12/2019*
  Thesis: *Nonlocal feature learning for image segmentation, facial video hallucination, and volumetric segmentation*
- **Xi'an Jiaotong University** — Xi'an, China
  Postgraduate Student - Pattern Recognition and Intelligent System; Advisor: Prof. Jianru Xue — *09/2013 - 06/2015*
  B.Eng. - Automation — *09/2009 - 07/2013*

Publication Summary

- **Journal Papers**: 9, including 6 in JCR Q1 journals and 6 with first/joint first/corresponding author.
- **Conference Papers**: 20, including 9 in CCF A conferences and 7 with first/joint first/corresponding author.
- **Patent Applications**: 2.

Sponsored Research Project Summary

- 5 projects with 360,000 CNY in total.

Major Subjects Taught

- Deep Learning (Undergraduate Level)
- Intelligent System (Undergraduate Level)

Supervisees/Co-supervisees

Summary	**Total**	**Completed**	**In Progress**
M.S. Students	7	1	6
B.S. Students	11	6	5

Major Service Activities

- Reviewer for around 20 journals or conferences, e.g. TIP, TMI, MIA, CVPR, ICCV, ECCV, AAAI, etc.
- Organizer of The 3rd International Workshop On Human-Centric Multimedia Analysis at ACM MM, 2022
- Thesis defense committee for postgraduate/undergraduate students, 2021-2022
- Committee for postgraduate reexamination, 2022
- Mentor for undergraduate students ($\sim$ 20), 2020-present
- Undergraduate admission advocate for XDU, 2021

Major Honors and Awards

- Champion of "Zero-shot Segmentation Challenge" of Zhejiang Lab Cup Global AI Competition, 2020
- HKU Postgraduate Scholarship, 2015-2019
- Excellent Graduate Student of Xi'an Jiaotong University, 2013

Skill Summary

- **Languages**: Python, C++
- **Packages**: PyTorch, TensorFlow, Keras, OpenCV, Scikit, Pytorch-Geometry
- **Tools**: MATLAB, GIT, Latex
- **Soft Skills**: Teaching, Writing, Event Management, Time Management

Peer-reviewed Conference Articles

∗ *indicates equal contribution, and* † *indicates corresponding author.*

1. Weizhi Zhong, **Chaowei Fang**, Yinqi Cai, Pengxu Wei, Gangming Zhao, Liang Lin, Guanbin Li. Identity-Preserving Talking Face Generation with Landmark and Appearance Priors. To Appear in IEEE International Conference on Computer Vision and Pattern Recognition (CVPR), 2023. **(CCF A, Acceptance Rate: 25.78%)**
2. Yuxiang Nie, **Chaowei Fang**, Lechao Cheng, Liang Lin, and Guanbin Li. Adapting Object Size Variance and Class Imbalance for Semi-Supervised Object Detection. To Appear in AAAI Conference on Artificial Intelligence, 2023. **(CCF A, Acceptance Rate: 19.6%)**
3. Kuo Wang, Jingyu Zhuang, Guanbin Li, **Chaowei Fang**, Lechao Cheng, Liang Lin, and Fan Zhou. De-biased Teacher: Rethinking IoU Matching for Semi-Supervised Object Detection. To Appear in AAAI Conference on Artificial Intelligence, 2023. **(CCF A, Acceptance Rate: 19.6%)**
4. Chengwei Pan, Baolian Qi, Gangming Zhao, Jiaheng Liu, **Chaowei Fang**, Dingwen Zhang, and Jinpeng Li. Deep 3D Vessel Segmentation based on Cross Transformer Network. IEEE International Conference on Bioinformatics and Biomedicine (BIBM), 2022.
5. Lechao Cheng, **Chaowei Fang**†, Dingwen Zhang, Guanbin Li, and Gang Huang. Compound Batch Normalization for Long-tailed Image Classification. ACM Conference on Multimedia, 2022. **(CCF A, Acceptance Rate: ∼25%)**
6. **Chaowei Fang**, Dingwen Zhang, Liang Wang, Yulun Zhang, Lechao Cheng, and Junwei Han. Cross-Modality High-Frequency Transformer for MR Image Super-Resolution. ACM Conference on Multimedia, 2022. **(CCF A, Acceptance Rate: ∼25%)**
7. Kuo Wang, Yuxiang Nie, **Chaowei Fang**, Chengzhi Han, Xuewen Wu, Xiaohui Wang, Liang Lin, Fan Zhou, and Guanbin Li. Double-Check Soft Teacher for Semi-Supervised Object Detection. International Joint Conference on Artificial Intelligence (IJCAI), 2022. **(CCF A, Acceptance Rate: ∼15%)**
8. **Chaowei Fang**∗, Liang Wang∗, Dingwen Zhang, Jun Xu, Yixuan Yuan, and Junwei Han. Incremental Cross-view Mutual Distillation for Self-supervised Medical CT Synthesis. IEEE International Conference on Computer Vision and Pattern Recognition (CVPR), 2022. **(CCF A, Acceptance Rate: ∼25.3%)**
9. Jiaming Li, **Chaowei Fang**†, and Guanbin Li. Gradient-Rebalanced Uncertainty Minimization for Cross-Site Adaptation of Medical Image Segmentation. Chinese Conference on Pattern Recognition and Computer Vision (PRCV), 2022.
10. Chengwei Pan∗, Gangming Zhao∗, Junjie Fang, Jiaheng Liu, **Chaowei Fang**, Dingwen Zhang, Jinpeng Li, and Yizhou Yu. Computer-aided Tuberculosis Diagnosis with Attribute Reasoning Assistance. MICCAI, 2022. **(Acceptance Rate: ∼30%)**
11. Xinkai Zhao, **Chaowei Fang**, De-Jun Fan, Xutao Lin, Feng Gao, and Guanbin Li. Cross-level Contrastive Learning and Consistency Constraint for Semi-supervised Medical Image Segmentation. IEEE International Symposium on Biomedical Imaging (ISBI), 2022
12. Junkai Huang∗, **Chaowei Fang**∗, Weikai Chen, Zhenhua Chai, Xiaolin Wei, Pengxu Wei, Liang Lin, and Guanbin Li. Trash to Treasure: Harvesting OOD Data with Cross-Modal Matching for Open-Set Semi-Supervised Learning. IEEE International Conference on Computer Vision (ICCV), 2021. **(CCF A, Acceptance Rate: ∼25.9%)**
13. Yiyang Huang, Xuefeng Liang, and **Chaowei Fang**. CALLip: Lipreading using Contrastive and Attribute Learning. ACM Conference on Multimedia, 2021. **(CCF A, Acceptance Rate: ∼27.9%)**
14. Baolian Qi, Gangming Zhao, Xin Wei, **Chaowei Fang**, Zhiqiang Chen, and Jinpeng Li. Weakly Supervised Disease Localization in Chest X-rays via Looking into Image Relations. IEEE International Conference on Bioinformatics and Biomedicine (BIBM), 2021.
15. Xinkai Zhao, **Chaowei Fang**, Feng Gao, De-Jun Fan, Xutao Lin, and Guanbin Li. Deep Transformers for Fast Small Intestine Grounding in Capsule Endoscope. IEEE International Symposium on Biomedical Imaging (ISBI), 2021.
16. Chang Cai, Xu Dou, **Chaowei Fang**, Meng Yang, and Zhongyu Li. Graph Neural Networks for the Cross-Domain Histopathological Image Classification. IEEE International Symposium on Biomedical Imaging (ISBI), 2021.
17. Jixin Wang, Sanping Zhou, **Chaowei Fang**, Le Wang, and Jinjun Wang. Meta Corrupted Pixels Mining for Medical Image Segmentation. International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI), 2020. **(Acceptance Rate: ∼30%)**
18. **Chaowei Fang**, Guanbin Li, Chengwei Pan, Yiming Li, and Yizhou Yu. Globally Guided Progressive Fusion Network for 3D Pancreas Segmentation. International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI), 2019. **(Acceptance Rate: ∼35%)**
19. Zhongyu Li∗, **Chaowei Fang**∗, and Shaoting Zhang. Deep Feature Representation for The Computational Analytics of 3D Neuronal Morphology. IEEE International Symposium on Biomedical Imaging (ISBI), 2018.
20. Yizhou Yu, **Chaowei Fang**, and Zicheng Liao. Piecewise Flat Embedding for Image Segmentation. IEEE International Conference on Computer Vision (ICCV), 2015. **(CCF A, Oral, Acceptance Rate: ∼3.3%)**
21. Zhongyu Li, Jihua Zhu, Ke Lan, Chen Li, and **Chaowei Fang**. Improved Techniques for Multi-view Registration with Motion Averaging. IEEE International Conference on 3D Vision, 2014.

Peer-reviewed Journal Articles

∗ *indicates equal contribution, and* † *indicates corresponding author.*

1. **Chaowei Fang**∗, Qian Wang∗, Lechao Cheng, Zhifan Gao, Chengwei Pan, Zhen Cao, Zhaohui Zheng, and Dingwen Zhang. Reliable Mutual Distillation for Medical Image Segmentation under Imperfect Annotations. IEEE Transactions on Medical Imaging, 2023. **(JCR Q1, IF2023: 11.037)**
2. Taiping Qu, Xiheng Wang, **Chaowei Fang**, Li Mao, Juan Li, Ping Li, Jinrong Qu, Xiuli Li, Huadan Xue, Yizhou Yu, and Zhengyu Jin. M3Net: A Multi-scale Multi-view Framework for Multi-phase Pancreas Segmentation Based on Cross-phase Non-local Attention. Medical Image Analysis, 2022. **(JCR Q1, IF2022: 13.828)**
3. Gangming Zhao∗, **Chaowei Fang**∗, Guanbin Li, Licheng Jiao, and Yizhou Yu. Contralaterally Enhanced Networks for Thoracic Disease Detection. IEEE Transactions on Medical Imaging, 2021. **(JCR Q1, IF2022: 11.037)**
4. Feida Zhu, **Chaowei Fang**†, and Kai-Kuang Ma. PNEN: Pyramid Non-Local Enhanced Networks. IEEE Transactions on Image Processing, 2020. **(CCF A, IF2022: 11.037)**
5. **Chaowei Fang**, Guanbin Li, Xiaoguang Han, and Yizhou Yu. Self-Enhanced Convolutional Network for Facial Video Hallucination. IEEE Transactions on Image Processing, 2019. **(CCF A, IF2022: 11.037)**
6. **Chaowei Fang**, Zicheng Liao, Yizhou Yu. Piecewise Flat Embedding for Image Segmentation. IEEE Transactions on Pattern Analysis and Machine Intelligence, 2018. **(CCF A, IF2022: 24.314)**
7. **Chaowei Fang**∗, Haibin Tian∗, Dingwen Zhang, Qiang Zhang, Jungong Han, and Junwei Han. Densely Nested Top-Down Flows for Salient Object Detection. SCIENCE CHINA Information Sciences, 2022. **(JCR Q1, IF2022: 7.275)**
8. Zhongyu Li, Xiayue Fan, Zengyi Shang, Lina Zhang, Haotian Zhen, and **Chaowei Fang**. Towards Computational Analytics of 3D Neuron Images using Deep Adversarial Learning. Neurocomputing, 2020.
9. Wei Zhang, **Chaowei Fang**, and Guanbin Li. Automatic Colorization with Improved Spatial Coherence and Boundary Localization. Journal of Computer Science and Technology, 2017.

Chinese Journal Articles

1. **Chaowei Fang**, Xue Li, Zhongyu Li, Licheng Jiao, and Dingwen Zhang. Interactive Dual-Model Learning for Semi-supervised Medical Image Segmentation. ACTA AUTOMATICA SINICA, 2022.

Chinese Patents

1. 程乐超, 李雪, 方超伟, 张鼎文. 一种基于双模型交互学习的半监督医学图像分割方法及装置. No. 202210228536.3. 05/17/2022.
2. 程乐超, 王良, 方超伟, 张鼎文. 一种跨模态核磁共振超分网络及图像超分辨率方法. No. 202210250052.9. 05/16/2022.

Sponsored Research Projects

- **国家自然科学青年基金**： 面向三维医学图像的语义分割及其域自适应算法研究，240,000 CNY, 2021-2023.
- **国际化种子基金**： 面向智能图像处理与理解的海外学术合作与交流，60,000 CNY, 2022-2024.
- **基本科研业务费**： 三维医学图像分割的领域自适应算法研究，30,000 CNY, 2021-2022.
- **华为智能基座课程建设项目**： 深度学习课程建设研究，20,000 CNY, 2022
- **中科川森**： 基于深度学习的表面划痕检测算法，100,000 CNY, 2020.

Research Experiences

- **Xidian University** — Xi'an, China
  Lecturer at School of Artificial Intelligence — *03/2020 - Current*
  1. Develop natural/medical image synthesis and restoration models.
  2. Develop network optimization algorithms under low-quality training data (few, noisy, or long-tailed training data).
  3. Develop learning algorithms for transferring models across domains.
  4. Design network architectures for combining multi-modal information.
- **The University of Hong Kong** — Hong Kong
  Ph.D. candidate at Department of Computer Science — *09/2015 - 12/2019*
  1. Develop pixel-wise non-linear embedding algorithm for unsupervised image segmentation.
  2. Develop video super-resolution models based on frame-recurrent propagation strategy and LSTM modules.
  3. Develop 3D medical image segmentation network for pancreas segmentation.
  4. Develop low-level image processing algorithms such as edge-aware smoothing and image colorization.
- **Deepwise** — Beijing, China
  Intern at Deepwise AI Lab — *02/2019 - 05/2019*
  1. Finish the research project on 3D medical image segmentation.
  2. Conduct engineering development on removing occlusions on medical images caused by manual marks.
- **Zhejiang University** — Hangzhou, China
  Visiting student at College of Computer Science and Technologies — *12/2014 - 05/2015, 06/2017 - 12/2017*
  1. Develop pixel-wise non-linear embedding algorithm for unsupervised image segmentation.
  2. Try to study on audio-driven face talking modelling.
- **Xi'an Jiaotong University** — Xi'an, China
  Postgraduate student at Institute of Artificial Intelligence and Robotics — *09/2013 - 06/2015*
  1. Maintain the lane generation system for an intelligent car.

Teaching Record

Semester	*Course Name*	*Course ID*	*#Student*	*Class Hour*	*Evaluation*
2022 Spring	Deep Learning	AI006007-01	80	8	-
2022 Spring	Specialty Experiment Based on Intelligent System	AI202009-02	86	32	-
2022 Spring	Deep Learning	AI205010-04	94	16	-
2022 Spring	Deep Learning	AI205010-05	39	16	-
2021 Spring	Deep Learning	AI006007-01	118	8	94.84
2021 Spring	Specialty Experiment Based on Intelligent System	AI202009-03	103	32	97.47
2020 Fall	Introduction to Deep Learning	AI204020-03	102	28	93.4

Professional Activities

- Organizer for Conferences/Symposiums
  1. Program Co-chair of The 3rd International Workshop On Human-Centric Multimedia Analysis at ACM MM 2022
- Reviewer for Journals
  1. IEEE Transactions on Image Processing (TIP)
  2. IEEE Transactions on Neural Networks and Learning Systems (TNNLS)
  3. IEEE Transactions on Cybernetics (TCyb)
  4. IEEE Transactions on Cogitive and Development Systems (TCDS)
  5. IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)
  6. IEEE Transactions on Multimedia (TMM)
  7. Medical Image Analysis (MIA)
  8. IEEE Journal of Biomedical and Health Informatics (JBHI)
  9. The Visual Computer
- Reviewer for Conferences
  1. CVPR 2021-2022
  2. AAAI 2021-2023
  3. MICCAI 2020, 2022
  4. ECCV 2022
  5. NeurIPS 2021
- Memberships
  1. 中国人工智能学会会员 (CAAI Member), 2022-2023
  2. 中国图像图形学会会员 (CSIG Member), 2021-2022
- Mentorships
  1. Mentor for undergraduate students (~ 20), 2020-present
- Committees
  1. Thesis defense committee for postgraduate students, 2021-2022
  2. Thesis defense committee for undergraduate students, 2021-2022
  3. Committee for postgraduate reexamination, 2022
  4. Secretary for postgraduate reexamination, 2020-2021
- Advocates
  1. Undergraduate admission advocate for XDU, 2021

Hobbies

- Reading professional/literary books; hiking; swimming.